

BOMBO HEADLAND, AustraliaWMO: **957490**Lat: **34.6533S**Lon: **150.8608E**Elev: **51**StdP: **14.67**Time Zone: **10.00 (AUS)**Period: **04-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	47.9	49.1	29.3	23.4	56.3	31.2	25.5	55.8	34.1	55.4	28.9	54.5	9.8	280	0.453

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.2	83.3	67.2	79.6	69.3	77.6	69.5	75.0	77.5	73.3	76.3	71.9	75.3	12.3	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.2	128.1	76.1	72.4	120.3	74.8	70.8	113.6	74.2	38.5	77.5	37.0	76.1	35.7	75.4	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.9	25.6	22.5	44.6	96.1	1.4	4.5	43.6	99.4	42.8	102.1	42.0	104.6	41.0	107.9		
(5)			38.6	76.5	0.9	3.7	38.0	79.1	37.5	81.3	37.0	83.4	36.4	86.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	64.0	71.2	70.8	69.4	65.8	61.5	57.8	56.6	57.4	60.4	63.2	66.1	68.4
	DBStd	6.23	3.62	3.43	3.26	3.45	3.46	2.75	3.12	3.41	4.31	4.68	4.02	3.62
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	1148	1	1	4	29	118	215	260	237	151	90	32	10
	CDD50	5116	657	581	602	474	357	235	206	230	313	408	483	570
	CDD65	791	193	163	141	53	10	0	1	2	14	34	65	115
	CDH74	1917	515	416	247	80	17	0	4	4	73	148	175	238
	CDH80	326	76	42	22	7	0	0	0	0	20	42	57	60
Wind	WSAvg	10.3	10.7	10.7	10.5	9.3	9.5	10.3	9.8	10.8	10.0	10.5	10.5	10.5
Precipitation	PrecAvg	50.0	4.7	5.1	6.5	5.1	4.8	5.2	2.0	3.1	3.0	3.7	4.3	3.5
	PrecMax	79.2	12.6	17.1	24.3	18.3	22.1	26.4	7.4	13.5	8.6	17.5	23.3	10.9
	PrecMin	23.7	0.7	0.3	0.5	0.3	0.1	0.0	0.0	0.1	0.1	0.2	0.3	0.3
	PrecStd	15.3	3.0	4.5	4.9	4.8	4.3	4.7	1.6	3.5	2.6	3.3	4.1	2.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	87.8	85.2	81.2	80.6	76.4	68.7	72.5	73.8	82.6	85.6	88.0	88.1
		MCWB	70.5	72.5	67.4	63.0	60.0	57.3	55.7	56.2	59.7	61.5	67.0	66.4
	2%	DB	80.4	80.0	78.0	75.9	72.0	66.2	67.2	69.0	74.4	77.7	78.0	78.9
		MCWB	73.0	72.9	69.6	65.9	61.2	57.3	55.2	56.3	57.8	62.1	64.7	66.8
	5%	DB	78.0	78.0	76.5	73.7	69.7	64.5	64.6	66.1	70.2	73.1	74.3	75.8
		MCWB	72.3	71.7	69.0	65.4	60.0	56.2	54.0	54.9	57.7	62.6	66.6	68.5
	10%	DB	76.5	76.4	75.0	71.9	67.6	62.9	62.6	64.0	67.3	70.5	72.3	74.1
		MCWB	71.2	70.6	68.3	64.7	59.1	55.3	53.1	54.1	58.1	62.4	65.6	67.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.7	77.2	74.2	70.6	65.4	62.7	61.5	60.8	63.7	68.2	70.9	73.5
		MCDB	80.1	79.8	76.0	74.5	69.8	64.4	66.0	66.5	70.3	72.4	77.4	76.0
	2%	WB	75.2	75.2	72.4	69.0	63.7	59.8	58.7	59.0	62.1	66.4	69.3	71.1
		MCDB	77.4	78.0	75.3	72.7	68.5	63.1	63.4	64.4	68.4	71.6	73.8	74.9
	5%	WB	73.4	73.5	70.9	67.4	62.0	58.4	56.5	57.5	61.0	65.2	68.2	69.9
		MCDB	76.5	76.6	74.7	71.6	67.6	61.9	61.1	63.1	67.3	70.3	72.7	74.0
	10%	WB	72.1	71.9	69.7	65.9	60.6	57.2	54.7	55.8	59.6	63.9	66.6	68.7
		MCDB	75.5	75.2	74.1	70.5	66.7	61.0	60.4	61.6	65.6	68.8	71.1	72.9
Mean Daily Temperature Range	5% DB	MDBR	9.2	8.9	9.1	10.0	10.5	9.3	11.2	11.4	12.3	11.6	10.5	9.8
		MCDBR	12.0	12.0	11.7	13.1	13.4	11.6	14.6	15.1	18.6	18.1	15.6	13.6
		MCWBR	7.1	7.2	7.6	8.2	8.5	7.4	8.8	9.3	9.7	10.1	8.5	7.4
	5% WB	MCDBR	11.1	10.5	10.0	10.9	11.5	8.7	11.8	12.9	14.8	13.4	12.3	10.9
		MCWBR	7.5	6.7	7.3	7.8	8.2	6.6	8.6	9.5	9.6	9.0	7.8	7.0
Clear-Sky Solar Irradiance	taub	0.397	0.386	0.363	0.334	0.317	0.307	0.298	0.307	0.339	0.355	0.385	0.397	
	taud	2.443	2.501	2.541	2.597	2.595	2.618	2.632	2.598	2.509	2.495	2.443	2.422	
	Ebn at Noon	298	295	290	281	269	264	274	287	293	301	300	299	
	Edh at Noon	38	35	31	26	23	21	22	26	31	35	38	39	
All-Sky Solar Radiation	RadAvg	2063	1752	1496	1202	939	732	862	1143	1541	1816	1938	2110	
	RadStd	155	152	122	101	56	68	69	62	92	143	166	159	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (33 neighbors)	+0.77	N/A	N/A	+1.35	N/A	N/A	-2	-153	+328	+167

Nomenclature: See separate page